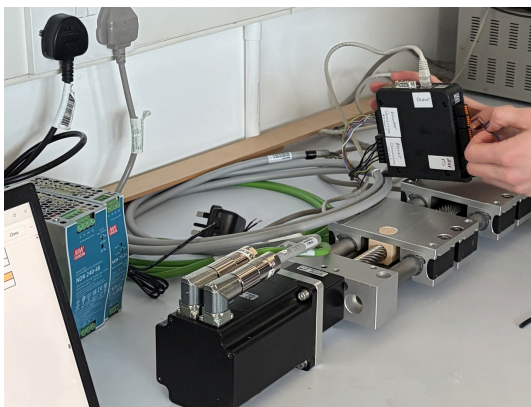


# Durham Heart Newsletter



## Finishing the final report

The deadline for the Heart Hackathon competition's final report is in early September, so we have been focusing on polishing and proofreading it, which will showcase all the work we have accomplished since the competition last year. This includes finalising our experimental results and sense-checking them to ensure our testing and computational modelling were sound. We have collected critical performance metrics, fluid dynamics data, and structural analysis figures to ensure our submission is backed by solid evidence. All the testing we have done allows us to demonstrate the viability of our total artificial heart.





## Mechanical Team

### Membrane design

We have been investigating the idea of using a folding polymer rather than an elastic membrane, since the lifespan should be extended if the material is not being stretched multiple times a minute. The main problems we anticipate are the possibility of trapped blood cells in the folds and the degradation of the material at the creases since there is likely to be stress concentrations.



## Electrical Team

### Testing the TET system

One of the major parts of our total artificial heart is the wireless TET system. It is essential in improving the quality of life and lowering the infection risk for patients. This is because it removes the need for a drive line (wires through the skin), which is notoriously problematic in existing Total Artificial Hearts. We focused on testing how to get the most power to the heart with the lowest losses from things like heat generation.



## Business Team

### Planning our trip to China

To compete in the Heart Hackathon finals held in Suzhou, China, we will be planning an international trip for multiple students to be able to attend the ISMCS conference. It will be an amazing opportunity to see emerging technologies in the industry, meet industry professionals and get to explore a different culture in the evenings.

# Thank you!